United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam Trimester: Spring 2026
Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (10 Marks) **[CO3]**

Global Descriptor Table	
Index	**Address**
F80 H	Base: 1F308810H, Limit: 78BD7H, Access: 54H, G = 1
9B2 H	Base: 05E4DBB8H, Limit: 3345AH, Access: 1FH, G = 0
D15 H	Base: 326125A3H, Limit: CC32FH, Access: 8AH, G = 0
073 H	Base: 00B2ACFEH, Limit: F56A9H, Access: 27H, G = 1
585 H	Base: 411310BDH, Limit: 50AEAH, Access: 53H, G = 0
B38 H	Base: 3CDF02F6H, Limit: C5072H, Access: 49H, G = 0

Local Descriptor Table	
Index	**Address**
F80 H	Base: 7A84718EH, Limit: 0FAE1H, Access: F9H, G = 0
9B2 H	Base: 2F72A2F3H, Limit: 29FDBH, Access: 0DH, G = 1
D15 H	Base: 10F3B41CH, Limit: 53027H, Access: B8H, G = 0
073 H	Base: 99A461B6H, Limit: C31ABH, Access: 62H, G = 0
585 H	Base: 056DC418H, Limit: E6357H, Access: 54H, G = 1
B38 H	Base: 8D1C994DH, Limit: 694CCH, Access: 14H, G = 1

Consider an imaginary microprocessor called Xion-32 that uses the **same Descriptor format** as the Intel 80386. The only difference is that the **G (Granularity) bit** in Xion-32 allows segment lengths to be specified in steps of **2 KB instead of 4 KB**. If the CS register contains **7C02H**, determine the following:

a.	Which descriptor table and entry will be selected?	**2**
	i. For finding the correct descriptor table → 1 mark ii. For finding the correct entry → 1 mark	
b.	Ending address of the selected segment.	**1**
	i. For using the correct formula → 0.5 marks ii. For finding the correct ending address → 0.5 marks	
c.	Is access to the segment granted, or not?	**1**
	i. For giving the correct answer of accessing the segment → 1 mark	
d.	What is the descriptor privilege level?	**1**
	i. For finding out the correct privilege level → 1 mark	
e.	Segment type (**CS/DS/SS/ES**).	**2**
	i. For finding the proper bits of segment type → 1 mark ii. For finding the correct segment → 1 mark	
f.	Calculate the maximum virtual memory size of Xion-32 in GB (GigaByte). **(Show necessary mathematical calculations.)**	**3**
	i. For finding total number of rows in the descriptor table → 1 mark	

	ii. For finding the size of each row → 1 mark iii. For finding total size of virtual memory → 1 mark	
Question 2: Answer all the questions. (10 Marks)		**[CO5]**
a.	For the command **analogWrite(x)**, Sajib is using a **duty cycle** with a **time period of 5s**. He sends the high pulse for **4s** and the low pulse for **1s**. He is using a **7-bit DAC system**. What is the value of **x**?	**3**
	i. For finding the duty cycle → 1 mark ii. For using the correct formula → 1 mark iii. For finding the correct value of x → 1 mark	
b.	Which type of communication protocol **(SPI or I2C)** is better suited for communication involving a small number of devices? While using the **I2C** communication protocol, a master device, **Master (index: 9H)**, sends **3-byte** data (char **'sPi'**) to a **slave device (index: 5H)**. Draw the corresponding sequence diagram. The ASCII codes for 'A' and 'a' are 65 and 97 respectively.	**1 + 3**
	i. For finding correct communication method → 1 mark ii. For correctly drawing the handshaking portion → 1 mark iii. For correctly drawing the data transfer portion → 1 mark iv. For correctly drawing the stop portion → 1 mark	
c.	A company wants to design a small wearable health-monitoring device that continuously measures body temperature and heart rate using sensors and sends an alert if abnormal readings are detected. The system can be built using either a **Raspberry Pi** or an **Arduino**. **Which platform would be more suitable and why?**	**3**
	i. For identifying the correct platform → 1 mark ii. For providing the correct reasons → 2 mark	
Question 3: Answer all the questions. (10 Marks)		**[CO3]**
a.	Consider the following **initial BIU fetch cycle pattern**: [Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch,Fetch,Fetch] Assume during execution of one instruction, the BIU fetches **three instructions** into the queue. The **2nd** instruction is a **'JUMP <5th instruction address>**, the **4th** instruction is a memory read operation, **7th** instruction is a '**MOV <address>'** instruction, **9th** instruction is '**JUMP <10th instruction address>. Draw the corresponding BIU and EU cycles.**	**3**
	i. For executing the first jump instruction correctly → 1 mark ii. For executing the 'MOV <address>' instruction correctly → 1 mark iii. For executing the second jump instruction correctly → 1 mark	
b.	A program has a virtual address space of **32 KB**. The system uses paging with a page size of **4 KB**. Currently, **5 pages** of the program are occupying primary memory. Assume that each page table entry is **8 bytes**. 1) **How many** pages are needed for the program? 2) **Calculate** the total page table size for this program. 3) **Calculate** the percentage of pages stored in **secondary storage.**	**1 + 1 + 2**
	i. For correctly finding the number of pages needed for the program → 1 mark ii. For correctly calculating the total page table size → 1 mark iii. For correctly calculating the number of pages stored in secondary memory → 1 mark iv. For correctly calculating the percentage → 1 mark	

c.	The 808x microprocessor is retrieving data from **RAM**. **Draw the timing diagram** for this operation showing the activities of Address-Data and Address-Status pins ($AD_0 - AD_{15}$, $A_{16}/S_3 - A_{19}/S_6$), $\overline{ALE}$, DEN, CLK, WR, $DT/\overline{R}$, $\overline{M}/IO$, $\overline{RD}$ in each clock cycle. **[Observe the pins carefully]**	**3**
	v. For correctly implementing T1 cycle → 1 mark vi. For correctly implementing T2 cycle → 0.5 marks vii. For correctly implementing T3 cycle → 1 mark viii. For correctly implementing T4 cycle → 0.5 marks	

Question 4: Answer all the questions. (10 Marks) **[CO4]**

RAM Content

Address	Content		Address	Content
0H			8H	1001 1010
1H			9H	0000 0101
2H			AH	0001 0000
3H			BH	1111 1010
4H			CH	0000 1010
5H			DH	0111 0111
6H			EH	0000 0011
7H			FH	1010 1010

Opcode Table

Mnemonic	Opcode
LDA	0011
ADD	0001
SUB	0010
OUT	1101
HLT	1011

Initially, the value of the **Accumulator Register is 5H, and the B Register is 10H.**

a.	Fill-up the **RAM content table** (0H - 7H) with the **machine code** of necessary instructions to show the output of the following expression (**in Hexadecimal**): $(-5 + 2 \times 9A - 3^2)_{16}$	**3**
	i. For correctly writing first three instructions → 1 mark ii. For correctly writing next three instructions → 1 mark iii. For correctly writing last two instructions → 1 mark	
b.	Determine the **Accumulator and B register** values after executing each instruction in [a].	**3**
	i. For correctly writing the values of AX → 1.5 marks ii. For correctly writing the values of BX → 1.5 marks	
c.	**CON =** $L_I \overline{E_I} L_A \overline{E_A}$ $\overline{E_U} \overline{S_U} L_O L_B$ $\overline{E_P} C_P \overline{L_M} \overline{CE}$ **[Observe the pins carefully]** **Currently,** the value of the **Memory Address Register** (**MAR)** is **9H, Accumulator Register is 5H, B Register** is **10H,** and **Output Register** is **7H.** The controller sequencer **accidentally** generates the following three control words in **three consecutive T-states (T4, T5 and T6)**: i) **7EAH**, ii) **4DBH**, iii) **72BH** Briefly describe the **operation** performed **in each T-state**. Also determine the contents of the **Accumulator (AX)**, **B Register**, and **Output Register** after each T-state using the **RAM content table provided above**. (**Hint:** The RAM outputs the contents of the memory location whose address is currently stored in the **MAR**).	**4**
	i. For correctly explaining the instruction → 1 mark ii. For correctly writing the values of AX → 1 mark iii. For correctly writing the values of BX → 1 mark iv. For correctly writing the values of Output Register → 1 mark	